

STATE BOARD OF TECHNICAL EDUCATION AND TRAINING
TELANGANA
DIPLOMA EXAMINATION (C-24)
C-24-SEP-2025
SEMESTER III , MID-I EXAM



CS-303
Data structures through 'C'

PCODE
243030

Exam Date: 03-09-2025

Session: FN

Duration: 1 Hour [09:30 AM To 10:30 AM]

[Total Marks: 20]

PART-A

Instructions:

1. Answer the following questions.
2. Each question carries **ONE** mark.

4 X 1 = 4

1. Define Pointer ?
2. What are array of pointers?
3. Which memory management function is used to deallocate the memory?
4. Define abstract data type.

PART-B

Instructions:

1. Answer the following questions.
2. Each question carries **THREE** marks.

2 X 3 = 6

- 5(a). Write about declaration and initialization of Pointers.
----- OR -----
- 5(b). Write a C program to swap two numbers using pointers.
- 6(a). Describe the Performance analysis of an algorithm.
----- OR -----
- 6(b). With a neat sketch describe the classification of Data structure?

PART-C

Instructions:

1. Answer the following questions.
2. Each question carries **FIVE** marks.

2 X 5 = 10

- 7(a). Explain with a C program to illustrate pointer variable as a member of structure.

104 7(b). Write a C program to sort an array of integer elements using pointers?

8(a). List out and briefly explain about Asymtotic notations?

----- OR -----

8(b). Write a C Program to demonstrate malloc() and free() function.